

Question set for Coursework 2

1. You're tasked to find the IAM (Identity & Access Management) users that accessed an AWS service in Frothly's AWS environment.

Refer to the following link to get an idea of what source type you need to query and what field in the results will have the answer you're seeking.

Link: <https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudtrail-log-file%20examples.html>

List out the IAM users that accessed an AWS service (successfully or unsuccessfully) in Frothly's AWS environment? Answer guidance: Comma separated without spaces, in alphabetical order. (Example: ajackson,mjones,tmiller)

Hint: Use aws:cloudtrail as the source type.

bstoll,btun,splunk_access,web_admin

The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk>enterprise' and various menu items like 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and a 'Find' search bar. Below this is a secondary navigation bar with 'Search', 'Analytics', 'Datasets', 'Reports', 'Alerts', and 'Dashboards'. The main content area is titled 'New Search' and contains a search bar with the following query:

```
index=botsv3 sourcetype=aws:cloudtrail
| search userIdentity.type="IAMUser"
| stats values(useridentity.userName) AS IAM_Users_List
| eval IAM_Users_List = mvsort(IAM_Users_List)
| eval IAM_Users_List=mvjoin(IAM_Users_List, ",")
```

 To the right of the search bar, there are buttons for 'Save As', 'Create Table View', and 'Close', and a 'Time range: All time' dropdown. Below the search bar, it shows '5,425 events (before 12/5/25 4:17:54.000 AM)' and a 'No Event Sampling' dropdown. There are also buttons for 'Job', 'Pause', 'Refresh', 'Download', and 'Smart Mode'. Below the search bar, there are tabs for 'Events', 'Patterns', 'Statistics (1)', and 'Visualization'. The 'Statistics (1)' tab is selected, showing a table with the following data:

IAM_Users_List
bstoll,btun,splunk_access,web_admin

2. The following links are provided to help you with this question.

Links:

- <https://aws.amazon.com/premiumsupport/knowledge-center/s3-bucketpublic-access/>
- <https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudwatchalarms-for-cloudtrail-additional-examples.html#cloudwatch-alarms-forcloudtrail-no-mfa-example>

Make sure you exclude events related to console logins.

It might be a good idea to do a keyword search query on this one. Don't forget to surround the keyword with asterisks.

What field would you use to alert that AWS API activity has occurred without MFA (multi-factor authentication)? Answer guidance: Provide the full JSON path. (Example: iceCream.flavors.traditional)

Hint: Use aws:cloudtrail as the source type.

userIdentity.sessionContext.attributes.mfaAuthenticated

The screenshot shows the Splunk Enterprise interface. At the top, there's a navigation bar with 'splunk>enterprise' and various menu items like 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', and 'Help'. Below this is a 'Search & Reporting' section with a search bar and a 'Search' button. The main area is titled 'New Search' and contains a search query: `index=botsv3 sourcetype=aws:cloudtrail | search (additionalEventData.MFAUsed=* OR userIdentity.sessionContext.attributes.mfaAuthenticated=*) | stats values(additionalEventData.MFAUsed) values(userIdentity.sessionContext.attributes.mfaAuthenticated) by eventType`. The search results show 2,159 events. The 'Statistics (2)' tab is selected, displaying a table with columns: 'eventType', 'values(additionalEventData.MFAUsed)', and 'values(userIdentity.sessionContext.attributes.mfaAuthenticated)'. The table has two rows: 'AwsApiCall' with 'false' and 'No', and 'AwsConsoleSignIn' with 'No'.

eventType	values(additionalEventData.MFAUsed)	values(userIdentity.sessionContext.attributes.mfaAuthenticated)
AwsApiCall	false	No
AwsConsoleSignIn	No	No

3. Look at the source types available in the dataset. There might be one in particular that holds information on hardware, such as processors.

What is the processor number used on the web servers? Answer guidance: Include any special characters/punctuation. (Example: The processor number for Intel Core i7-8650U is i7-8650U.)

Hint: Use hardware as the source type in Splunk Search for find hardware information such as CPU statistics, hard drives, network interface cards, memory, and more.

E5-2676

The screenshot shows the Splunk Search interface with a search query `index=botsv3 sourcetype=hardware` and a time range of "All time". The search results show 3 events. The first event is expanded, showing a table of hardware information for a CPU E5-2676 v3. The table lists various hardware components and their values.

KEY	VALUE
CPU_TYPE	Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz
CPU_CACHE	30720 KB
CPU_COUNT	2
HARD_DRIVES	xvda 8 GB;
NIC_TYPE	<notAvailable>
NIC_COUNT	1
MEMORY_REAL	4841808 kB
MEMORY_SWAP	0 kB

Below the table, the host is identified as `gacruX.F09cbc261e84259b54` with source `hardware` and sourcetype `hardware`.

4. Questions 4-6:

A common misconfiguration involving AWS is publically accessible S3 buckets. Read the following resource to understand ACLs and S3 buckets.

Link: https://docs.aws.amazon.com/AmazonS3/latest/API/API_PutBucketAcl.html

Question 4: Bud accidentally makes an S3 bucket publicly accessible. What is the event ID of the API call that enabled public access? Answer guidance: Include any special characters/punctuation.

Hint: Use aws:cloudtrail as the source type to search for the PutBucketAcl event.

ab45689d-69cd-41e7-8705-5350402cf7ac

New Search [Save As] [Create Table View] [Close]

index=botsv3 sourcetype=aws:cloudtrail eventName="PutBucketAcl"
| search userIdentity.userName="bstoll"
| table _time eventName eventID userIdentity.userName requestParameters.* responseElements.*
| sort _time ascending

Time range: All time [Q]

✓ 2 events (before 12/6/25 3:05:39.000 AM) [No Event Sampling] [Job] [II] [≡] [↗] [⬇] [⬆] [Smart Mode]

Events Patterns Statistics (2) Visualization

Show: 100 Per Page [Format] [Preview: On]

_time	eventName	eventID	userIdentity.userName	requestParameters.AccessControlPolicy.AccessControlList.Grant[0].Grantee.DisplayName	requestParameters.AccessControlPolicy.AccessControlList.Grant[0].Grantee.Type
2018-08-20 21:01:46	PutBucketAcl	ab45689d-69cd-41e7-8705-5350402cf7ac	bstoll	bstoll	bstoll
2018-08-20 21:57:54	PutBucketAcl	9a33d8df-1e16-4d58-b36d-8e80ce68f8a3	bstoll	bstoll	bstoll

5. What is Bud's username?

bstoll

New Search [Save As] [Create Table View] [Close]

index=botsv3 sourcetype=aws:cloudtrail eventName="PutBucketAcl"
| search userIdentity.userName="bstoll"
| table _time eventName eventID userIdentity.userName requestParameters.* responseElements.*
| sort _time ascending

Time range: All time [Q]

✓ 2 events (before 12/6/25 3:05:39.000 AM) [No Event Sampling] [Job] [II] [≡] [↗] [⬇] [⬆] [Smart Mode]

Events Patterns Statistics (2) Visualization

Show: 100 Per Page [Format] [Preview: On]

_time	eventName	eventID	userIdentity.userName	requestParameters.AccessControlPolicy.AccessControlList.Grant[0].Grantee.DisplayName	requestParameters.AccessControlPolicy.AccessControlList.Grant[0].Grantee.Type
2018-08-20 21:01:46	PutBucketAcl	ab45689d-69cd-41e7-8705-5350402cf7ac	bstoll	bstoll	bstoll
2018-08-20 21:57:54	PutBucketAcl	9a33d8df-1e16-4d58-b36d-8e80ce68f8a3	bstoll	bstoll	bstoll

6. What is the name of the S3 bucket that was made publicly accessible?

Hint: Use aws:cloudtrail as the source type.

frothlywebcode

New Search Save As Create Table View Close

```
index=botsv3 sourcetype=aws:cloudtrail eventName="PutBucketAcl"
| search userIdentity.userName="bstoll"
| table _time eventName eventID userIdentity.userName requestParameters.* responseElements.*
| sort _time ascending
```

Time range: All time Q

✓ 2 events (before 12/6/25 4:02:57.000 AM) No Event Sampling Job || ■ ↶ ↷ ⬇ ⬆ Smart Mode

Events Patterns Statistics (2) Visualization

Show: 100 Per Page Format Preview: On

Policy.Owner.DisplayName	requestParameters.AccessControlPolicy.Owner.ID	requestParameters.AccessControlPolicy.xmlns	requestParameters.ac[]	requestParameters.bucketName
	4c018053e740f45beb45f68c0f5eff6347745488ae540130432c9fc64fae310d	http://s3.amazonaws.com/doc/2006-03-01/		frothlywebcode

7. You're tasked with identifying a text file uploaded to the S3 bucket. Here is a link for more information related to this topic.

Link: https://docs.aws.amazon.com/AmazonS3/latest/API/API_PutObject.html

Since you know the name of the S3 bucket, you should easily find the answer to this question.

You will need to query a different AWS-related source type. HTTP status code might be helpful as well.

What is the name of the text file that was successfully uploaded into the S3 bucket while it was publicly accessible? Answer guidance: Provide just the file name and extension, not the full path. (Example: filename.docx instead of /mylogs/web/filename.docx)

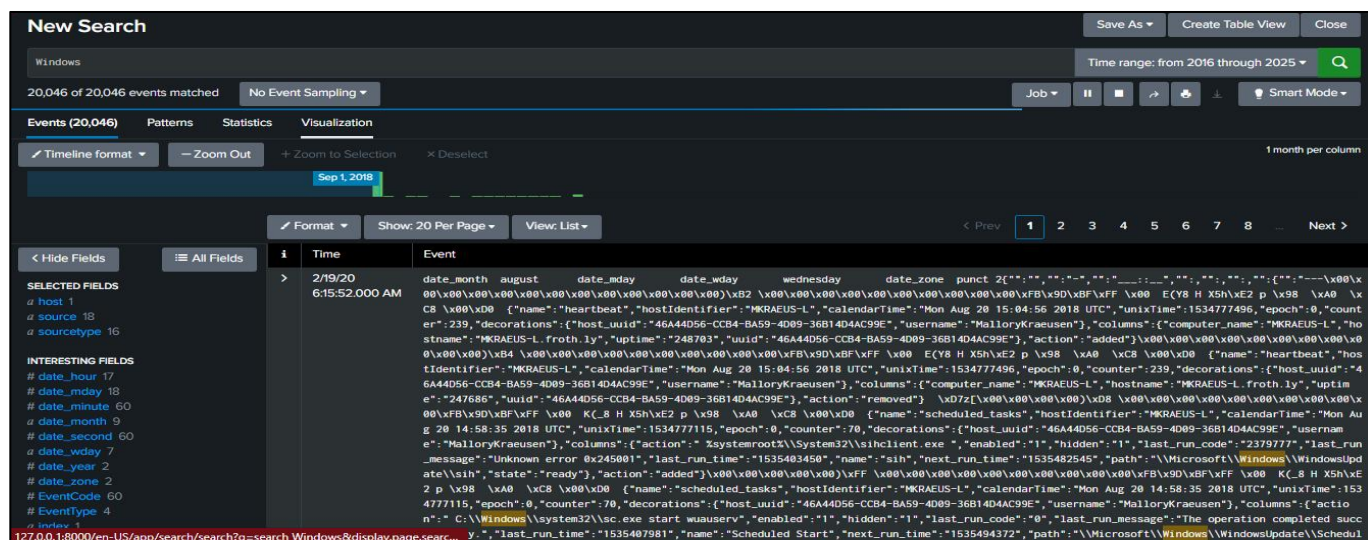
Hint: Use aws:s3:accesslogs.

OPEN_BUCKET_PLEASE_FIX.txt

The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk>enterprise' and various menu items like 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a 'Search & Reporting' section with tabs for 'Search', 'Analytics', 'Datasets', 'Reports', 'Alerts', and 'Dashboards'. The 'Search' tab is active, showing a 'New Search' page. The search query is: `index=botsv3 sourcetype=aws:s3:accesslogs frothlywebcode REST.PUT.OBJECT "*.txt"`. Below the query, there are two lines of SPL: `| rex field=_raw "PUT \/(?<FileName>[^\s]+)\sHTTP\S+\s(?<Status>\d+)"` and `| table _time, FileName, Status`. The search results show 1 event (before 12/6/25 7:40:48.000 AM). The event details are: `_time`: 2018-08-20 21:02:44, `FileName`: OPEN_BUCKET_PLEASE_FIX.txt, and `Status`: 200.

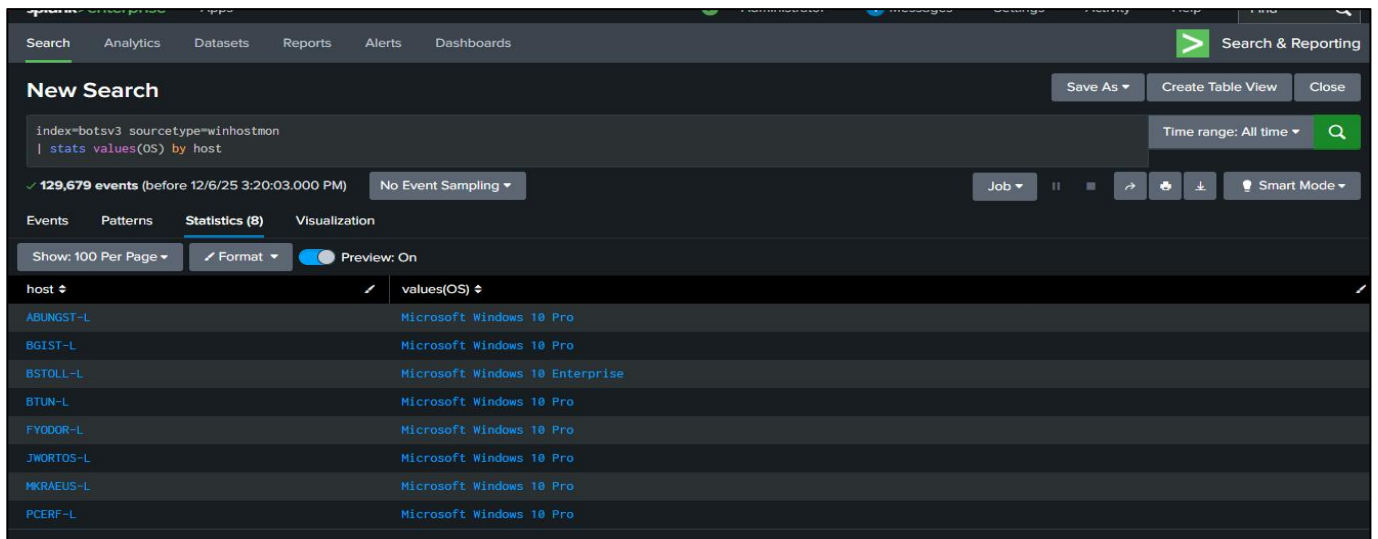
_time	FileName	Status
2018-08-20 21:02:44	OPEN_BUCKET_PLEASE_FIX.txt	200

winhostmon, Windows, OS



One of the fields within this source type clearly has the answer, but which is it?

OS



The screenshot shows the Splunk Search interface with the following details:

- Search Bar:** `index=botsv3 sourcetype=winhostmon | stats values(OS) by host`
- Time Range:** All time
- Results:** 129,679 events (before 12/6/25 3:20:03.000 PM)
- Table View:** Shows columns for `host` and `values(OS)`.

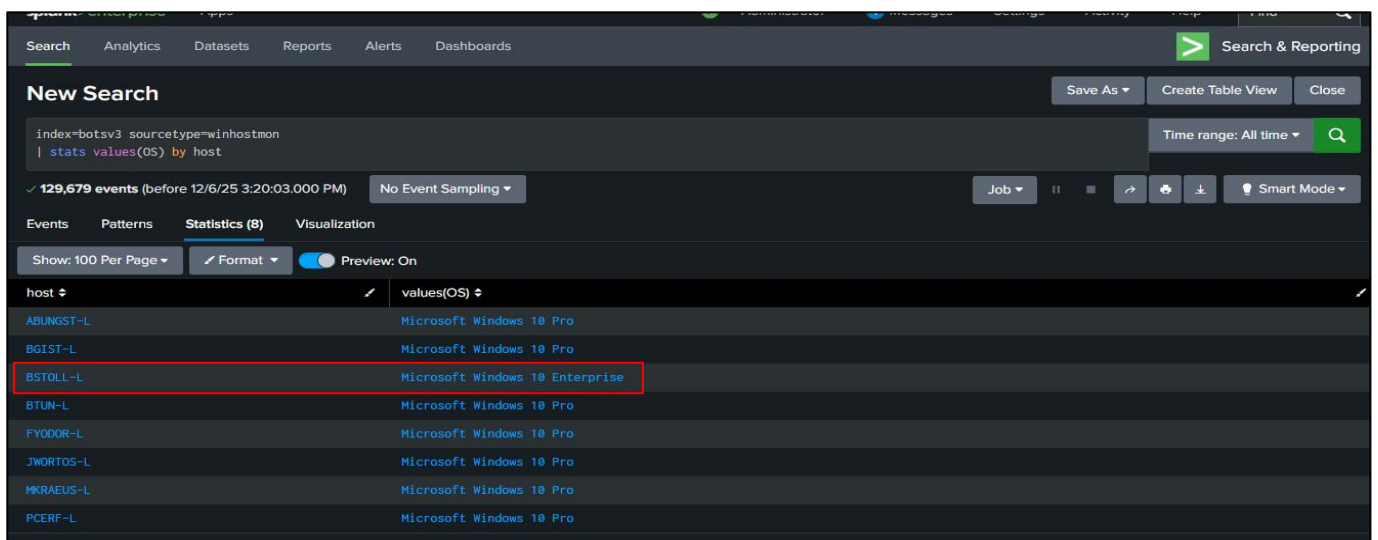
host	values(OS)
ABUNGST-L	Microsoft Windows 10 Pro
BGIST-L	Microsoft Windows 10 Pro
BSTOLL-L	Microsoft Windows 10 Enterprise
BTUN-L	Microsoft Windows 10 Pro
FYODOR-L	Microsoft Windows 10 Pro
JWORTOS-L	Microsoft Windows 10 Pro
MKRAEUS-L	Microsoft Windows 10 Pro
PCERF-L	Microsoft Windows 10 Pro

Perhaps expanding upon your search to count on the operating systems and hosts will be helpful.

What is the FQDN of the endpoint that is running a different Windows operating system edition than the others?

Hint: Start with winhostmon as the source type.

BSTOLL-L.froth.ly



This screenshot is identical to the one above, but with the row for `BSTOLL-L` highlighted with a red rectangle, indicating it is the correct answer to the question.

host	values(OS)
ABUNGST-L	Microsoft Windows 10 Pro
BGIST-L	Microsoft Windows 10 Pro
BSTOLL-L	Microsoft Windows 10 Enterprise
BTUN-L	Microsoft Windows 10 Pro
FYODOR-L	Microsoft Windows 10 Pro
JWORTOS-L	Microsoft Windows 10 Pro
MKRAEUS-L	Microsoft Windows 10 Pro
PCERF-L	Microsoft Windows 10 Pro

SearchAnalyticsDatasetsReportsAlertsDashboards

Search & Reporting

New Search

Save AsCreate Table ViewClose

Time range: All time

```
index=botsv3 sourcetype=WinEventLog:*
| rex field=_raw "ComputerName=(?<ComputerName>[^\s,;]+)"
| eval host_fqdn = coalesce(ComputerName, host)
| dedup host_fqdn
| table host_fqdn
```

8 of 6,490 events matchedNo Event SamplingJobPauseStopRefreshSmart Mode

EventsPatternsStatistics (8)Visualization

Show: 100 Per PageFormatPreview: On

host_fqdn

BGIST-L.froth.ly

FYODOR-L.froth.ly

BSTOLL-L.froth.ly

PCERF-L.froth.ly

MKRAEUS-L.froth.ly

BTUN-L.froth.ly

JWORTOS-L.froth.ly

SEPM